a) i) primary data: data collected from original sources ...

Secondary data: compiled by others but use for our own study ...

Sampling: subgroup of the population, method include random sampling + stratified random sampling, sample should be representative of the populat and not introduce bias ...

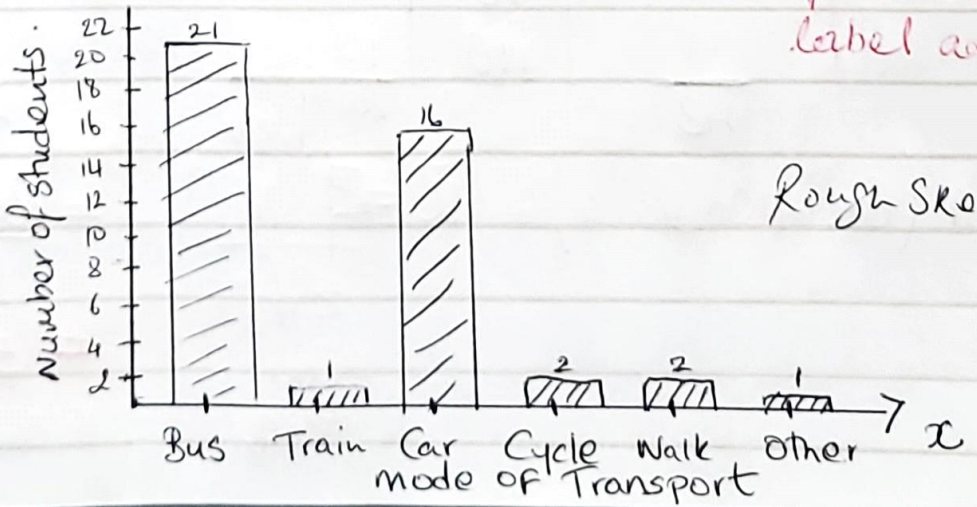
(expand on these, see notes).

ii) primary data - collected by the tutor via class survey and used for the tutor's own analysis ...

iii) categorical data - finite number / discre number OR modes of transport ...

iv) categorical data therefore the / an appropri graphical representation is a bar chart.

space between ba
label axis ...



Rough Sketch!

b) i) high positive correlation - in general the more time spent studying, the higher the grade achieved in the exam. i.e. as the value on the x-axis increases, so to just the value on the y-axis.

ii) A exception - data value/point that deviates/is away from the main trend/cluster of points [In figure 3.1 student studied ~ 6 hours (much less than others) but achieved the highest mark %. in the exam] exception. In general students that studied for 5/6 hours achieved lower marks %. in the exam.

iii) predicted exam mark = ~60%
read this value from the graph using trend line.

c)

Time	Frequency	M_i	$f_i M_i$	M_i^2	$f_i M_i^2$
0-20	9	10	90	100	900
20-40	9	30	270	900	8,100
40-60	7	50	350	2500	17,500
60-80	4	70	280	4900	19,600
80-100	3	90	270	8100	24,300
100-120	11	110	1210	12,100	133,100
	43		2470		203,500

(Σ over $f_i M_i$; Σ over $f_i M_i^2$; Σf_i = sum freq)

Mean (Grouped Data): $\bar{x} = \frac{\sum f_i M_i}{\sum f_i} = \frac{2470}{43}$ (sum freq, midpoint)

$= 57.44$ to 2 decimal places

ii) Standard Deviation (Grouped Data)

use $s^2 = \frac{\sum f_i M_i^2 - M(\bar{x})^2}{M-1}$ — total freq Σf_i

$$= \frac{203,500 - 43(57.44)^2}{43-1}$$

$$= \frac{203,500 - 141,872.2048}{42}$$

$s^2 = 1467.33$

$s = \sqrt{1467.33}$

$s = 38.31$ to 2 decimal places.

OR using

$$s = \sqrt{\frac{\Sigma f_i (M_i - \bar{x})^2}{\Sigma f_i - 1}}$$

		Σ
$\Rightarrow$	$9 \times (10 - 57.44)^2$	$= 20,254.9824$
	$4 \times (30 - 57.44)^2$	$= 6,776.5824$
	$7 \times (50 - 57.44)^2$	$= 387.4752$
	$4 \times (70 - 57.44)^2$	$= 631.0144$
	$3 \times (90 - 57.44)^2$	$= 3,180.4608$
	$11 \times (100 - 57.44)^2$	$= 30,388.0896$
		$61,618.6048$

$s = \sqrt{\frac{61,618.6048}{42}}$ $\leftarrow$ $\frac{1467.11}{42}$

$= 38.31$

d) 4 letter codes, first 12 letters of alphabet, no letter can be Repeated

$$12 \times 11 \times 10 \times 9 = 11880$$

OR

$${}^{12}P_4 = \frac{12!}{(12-4)!} = 11880.$$

e) $P[H < 167]$ $\mu = 165$, $\sigma = 4.5$

$$a = \frac{167 - 165}{4.5} = 0.44$$

$$P[H<167] = P[z < 0.44] = 1 - P[z > 0.44]$$

look up on z table.

$$= 1 - 0.3300$$

$$P[H<167] = 0.6700 \text{ or } \sim 67\%$$